

The empirical recurrence for OEIS A264503, recovered from its tail

Adrian Perez Fontelles

Independent researcher

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Abstract

OEIS A264503 records “Empirical recurrence of order 84” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 4096$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the whole sequence, so the recurrence holds from $n = 85$ — every index at which it can be written down. Its last coefficient is $c_{84} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-84 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A264503 is “Number of $(n+1)X(5+1)$ arrays of permutations of $0..n*6+5$ with each element having directed index change -1,1 0,-1 0,1 or 1,0.” It has offset 1 and begins

12, 137, 1235, 11464, 110185, 1041411, 9813822, 92711566, ...

The entry states:

Empirical recurrence of order 84 (see link above)

As of the “Last modified” line on the live entry (Nov 15 2015 18:35:54, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 4096$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 4096$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Here the stated order is already the minimal order of the sequence itself, so no tail has to be taken.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 8192$ exact terms from $n = 1$ on, it returns order 84 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{84} c_i a(n-i),$$

c_1	7	c_{22}	−1635725779	c_{43}	593587672787	c_{64}	−1116984026
c_2	13	c_{23}	1932756755	c_{44}	−642026025291	c_{65}	792626528
c_3	177	c_{24}	−2484426677	c_{45}	449427095243	c_{66}	−318380127
c_4	−500	c_{25}	8156459937	c_{46}	−517524020961	c_{67}	148708117
c_5	−1619	c_{26}	−9804100476	c_{47}	526338996198	c_{68}	−99631404
c_6	−11053	c_{27}	11120929215	c_{48}	−346093768325	c_{69}	36019996
c_7	7214	c_{28}	−30506220811	c_{49}	353456515146	c_{70}	−14042644
c_8	44766	c_{29}	38326053314	c_{50}	−337942934955	c_{71}	8904498
c_9	271556	c_{30}	−39090939590	c_{51}	207815460961	c_{72}	−2876396
c_{10}	−46411	c_{31}	89284505851	c_{52}	−188746952475	c_{73}	903978
c_{11}	−489561	c_{32}	−114413359152	c_{53}	169756824628	c_{74}	−545255
c_{12}	−3327084	c_{33}	105624099050	c_{54}	−97090258256	c_{75}	156282
c_{13}	903680	c_{34}	−206318021327	c_{55}	78418527967	c_{76}	−37313
c_{14}	1193563	c_{35}	260333930107	c_{56}	−66403904170	c_{77}	21593
c_{15}	23077030	c_{36}	−220876375605	c_{57}	35109413945	c_{78}	−5466
c_{16}	−19304680	c_{37}	374818376023	c_{58}	−25126768893	c_{79}	882
c_{17}	23314059	c_{38}	−454122368611	c_{59}	20049354497	c_{80}	−496
c_{18}	−109888151	c_{39}	359249622625	c_{60}	−9737308997	c_{81}	111
c_{19}	226482988	c_{40}	−533219034104	c_{61}	6127665796	c_{82}	−9
c_{20}	−285117109	c_{41}	612143763701	c_{62}	−4609868692	c_{83}	5
c_{21}	499801202	c_{42}	−455141045897	c_{63}	2042976246	c_{84}	−1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 85$, and it is the recurrence OEIS A264503 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 1$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{84} - c_1 x^{84-1} - \dots - c_{84}$. Its constant term is $-c_{84} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 84 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 84, so $\deg q = 84$, and it is monic. It holds from some index t on. From $\max(t, 1)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 84, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 84, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-84 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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